

CASA GRANDE, AZ, USA

WMO: 722748

Lat: 32.950N

Lon: 111.767W

Elev: 1462

StdP: 13.94

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.7	35.4	3.1	6.8	45.0	8.6	9.0	50.4	22.8	58.2	19.7	59.5	2.7	50	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	108.5	69.7	106.5	69.2	104.9	68.8	74.5	92.7	73.6	93.5	72.9	93.4	8.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	118.4	80.0	68.9	112.0	81.1	66.3	102.2	83.0	39.0	92.9	38.2	93.0	37.4	93.0	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
20.7	17.9	15.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			DB	26.2	112.3	2.8	2.2	24.2	113.9	22.5	115.1	21.0	116.3	18.9	117.9	
			WB	21.6	76.0	3.4	1.2	19.2	76.8	17.1	77.5	15.2	78.2	12.7	79.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	70.9	52.5	55.0	61.8	68.1	76.8	86.4	91.2	90.0	84.4	71.9	60.2	51.5
	DBStd	15.33	5.80	6.15	6.43	6.90	6.46	5.36	4.13	4.04	4.85	7.02	6.53	5.60
	HDD50	120	40	20	2	0	0	0	0	0	0	0	7	51
	HDD65	1466	389	281	138	47	3	0	0	0	0	22	169	417
	CDD50	7750	117	161	368	544	831	1093	1278	1239	1031	678	312	98
	CDD65	3623	1	2	40	140	369	643	813	774	581	236	24	0
	CDH74	56472	22	114	836	2283	5715	10746	13255	12119	8067	2869	436	10
CDH80	35261	0	9	244	1023	3337	7319	9049	7968	4846	1366	100	0	
Wind	WSAvg	5.5	4.7	4.9	5.3	6.4	6.5	6.1	6.4	5.7	5.4	5.1	4.6	4.4
Precipitation	PrecAvg	8.7	0.7	0.7	0.9	0.3	0.1	0.1	0.9	1.9	0.8	0.8	0.7	1.2
	PrecMax	15.2	3.0	2.2	3.1	1.5	0.6	0.9	3.3	6.2	2.0	3.7	2.4	4.7
	PrecMin	4.9	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0
	PrecStd	2.8	0.7	0.7	0.8	0.4	0.2	0.2	0.8	1.6	0.7	0.9	0.7	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.9	81.4	90.4	97.1	105.7	109.4	111.3	109.4	106.0	99.3	87.9	75.1
		MCWB	52.5	54.3	58.6	61.3	65.9	68.0	70.8	70.4	69.2	64.9	58.6	52.3
	2%	DB	72.6	78.5	85.8	92.9	100.4	108.0	108.8	107.9	102.3	95.4	82.4	72.0
		MCWB	51.2	53.4	57.1	59.2	63.2	67.8	70.4	70.7	68.4	63.8	55.9	52.0
	5%	DB	69.9	73.3	82.1	90.1	98.8	106.1	107.4	105.7	100.2	91.1	79.4	67.8
		MCWB	50.3	51.5	55.2	58.2	62.5	67.0	70.1	70.9	67.9	61.7	55.0	50.4
	10%	DB	65.8	70.0	79.2	86.4	95.1	103.6	104.4	102.1	98.8	88.2	75.0	64.0
		MCWB	48.9	50.6	53.9	56.8	60.5	66.0	69.8	70.5	67.6	61.1	53.0	48.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.9	57.4	60.1	62.1	66.4	71.9	76.0	75.8	74.3	70.8	60.4	57.4
		MCDB	65.4	70.4	84.9	93.5	101.2	98.5	90.8	92.5	90.2	81.0	76.3	66.6
	2%	WB	54.3	55.5	58.0	60.2	64.4	70.3	74.7	74.6	73.0	67.5	58.2	54.8
		MCDB	65.4	69.6	80.9	90.7	97.9	98.4	92.4	93.3	89.8	87.1	75.7	64.8
	5%	WB	52.4	53.7	56.4	58.6	62.9	68.7	73.7	73.7	71.9	65.1	56.4	52.4
		MCDB	64.7	68.8	78.6	87.7	95.6	98.7	94.0	93.7	89.4	84.7	74.7	63.4
	10%	WB	50.4	52.0	54.9	57.2	61.6	67.3	72.9	73.0	70.9	62.9	54.8	50.4
		MCDB	63.2	67.2	76.2	84.7	93.4	99.6	94.3	93.7	89.2	83.7	73.2	61.9
Mean Daily Temperature Range	5% DB	MDBR	25.3	25.6	28.6	30.0	30.9	31.7	24.4	23.5	25.3	27.5	27.0	24.4
		MCDBR	30.1	31.9	33.0	34.5	34.2	32.9	28.4	27.1	27.8	31.2	31.5	29.3
		MCWBR	16.2	16.0	15.4	15.8	15.5	13.9	9.8	8.2	9.8	13.4	16.0	15.9
	5% WB	MCDBR	23.7	26.8	30.4	32.4	32.4	27.6	21.9	22.1	21.2	24.5	27.2	23.8
		MCWBR	13.3	13.8	13.9	14.9	14.4	10.5	6.2	6.1	7.1	10.7	14.3	13.4
Clear-Sky Solar Irradiance	taub		0.275	0.280	0.305	0.314	0.322	0.336	0.396	0.383	0.342	0.314	0.285	0.275
	taud		2.563	2.556	2.461	2.427	2.434	2.400	2.360	2.386	2.469	2.495	2.533	2.562
	Ebn at Noon		304	312	310	309	305	299	281	284	293	295	297	298
	Edh at Noon		25	28	34	36	36	38	39	37	33	30	26	24
All-Sky Solar Radiation	RadAvg		1101	1405	1878	2322	2593	2665	2310	2139	1928	1581	1218	1001
	RadStd		78	105	84	62	86	110	108	67	76	90	54	48

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-151	+196	N/A

Nomenclature: See separate page